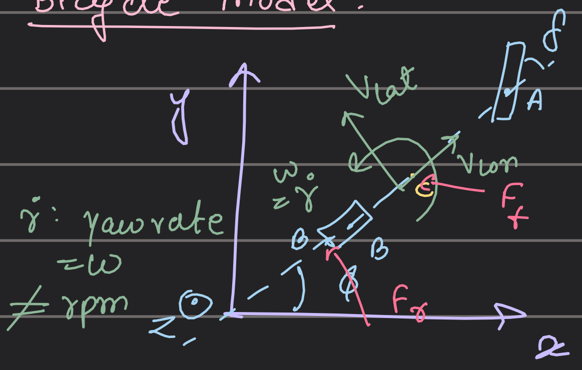
SLAM-Checkpoint-2

Bicycle Model:



$BC = l_r$

$AC = l_f$

Assumptions:

- wheels of the front and rear axle are lumped together into one front wheel & one rear wheel.

- reference point c is the vehicle's centre of gravity
- motion of the vehicle happens only in the x-y plane

- $l = l_f + l_r$
  - base length (seperation b/w both the tires)

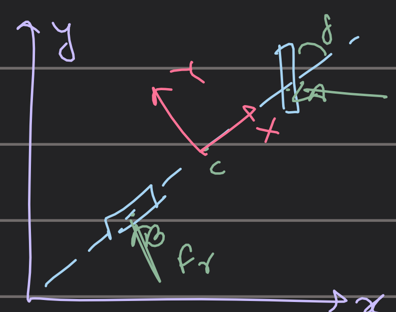
- only the lateral tire forces are generated by a linear tire model

- $C_{\alpha f}, C_{\alpha r}$: cornering stiffness of the combined front & rear tires
- δ: steering angle , ϕ : head angle
- assuming the steering angle to be small, $\sin\delta = \delta = \tan\delta$, $\cos\delta = 1$

- the longitudinal vel : V_{lon} is const

$$\boxed{\begin{aligned} \dot{x} &= v\cos\theta \, dt \\ \dot{y} &= v\sin\theta \, dt \\ \dot{\theta} &= \omega dt \end{aligned}} \Rightarrow \text{Simple Bicycle model}$$

Laws of motion:



Lateral dynamics:

$$m a_y(x,y) = \sum_i F_{y_i}$$

$$= F_r + F_f \cancel{\cos\delta} \nearrow 1$$

$$= F_r + F_f$$

car frame $\downarrow$ non-inertial

$$a_y(x,y) = \dot{v}_{Lat} + \omega V_{lon}$$

$+\omega V_{lon}$ → centripital acc^n

$$m(\dot{v}_{Lat} + \omega V_{lon}) = F_r + F_f \quad \text{— (i)}$$

Yaw Dynamics:

moments

Torque eqn about C: $I_z \dot{\omega} = \sum_i M_i^{(c)} = -F_r L_r + L_r F_f \cos\delta = -F_r L_r + L_r F_f$ — (2)

Tire forces:

cornering stiffness

Linear tire model: $F_f = -C_{\alpha,f}\,\alpha_f$, $\quad F_r = -C_{\alpha,r}\,\alpha_r$

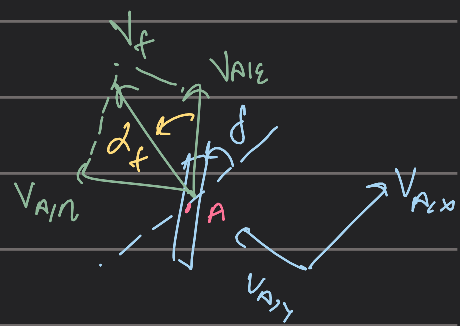
slip angle

rear tire

velocity geometry: $\tan\alpha_f \approx \alpha_f = \dfrac{V_{A,n}}{V_{A,\varepsilon}}$

$$\tan\alpha_r \approx \alpha_r = \frac{V_{B,y}}{V_{B,x}}$$

front tire



Kinematics: $V_{B,x} = V_{lon}$, $\quad V_{B,y} = V_{lat} - \omega L_r$

$V_{A,x} = V_{lon}$, $\quad V_{A,y} = V_{lat} + \omega L_f$

$$V_{A,\varepsilon} = V_{A,x}\cos\delta + V_{A,y}\sin\delta = V_{A,x} + V_{A,y}\,\delta$$

$$V_{A,n} = V_{A,y}\cos\delta - V_{A,x}\sin\delta = V_{A,y} - V_{A,x}\,\delta$$

$$F_f = -C_{\alpha,f}\,\alpha_f = -C_{\alpha,f}\frac{V_{A,n}}{V_{A,\varepsilon}} = -C_{\alpha,f}\left(\frac{-V_{lon}\delta + V_{lat} + \omega L_f}{V_{lon} + (V_{lat} + \omega L_f)\delta}\right)$$

$$(V_{lon} \gg (V_{lat} + \omega L_f)\delta) \quad \approx \quad -C_{\alpha,f}\,\delta - \frac{C_{\alpha,f}\,V_{lat} + \omega L_f}{V_{lon}} \qquad \text{— (3)}$$

$$F_r = -C_{\alpha,r}\,\alpha_r = -C_{\alpha,r}\frac{V_{B,y}}{V_{B,x}} \approx -C_{\alpha,r}\frac{V_{lat} - \omega L_r}{V_{lon}} \qquad \text{— (4)}$$

State space representation: Subs (3) & (4) in (1)

$$m(\dot{V}_{lat} + \omega V_{lon}) = -C_{\alpha,r}\frac{V_{lat} - \omega L_r}{V_{lon}} + C_{\alpha,f}\,\delta - C_{\alpha,f}\frac{V_{lat} + \omega L_f}{V_{lon}}$$

$$\dot{V}_{lat} = -\frac{C_{\alpha r} + C_{\alpha f}}{m V_{lon}} V_{lat} + \frac{C_{\alpha,r}L_r - C_{\alpha,f}L_f}{m V_{lon}}\,\omega - V_{lon}\,\omega + \frac{C_{\alpha f}}{m}\,\delta \qquad \text{— (A)}$$

Subs (3) & (4) in (2)

$$I_z \dot{\omega} = -l_r \left(-C_{\alpha,r} \frac{V_{lat} - \omega l_r}{V_{lon}}\right) + l_f \left(C_{\alpha,f}\, \delta - C_{\alpha f} \frac{V_{lat} + \omega l_f}{V_{lon}}\right)$$

$$\dot{\omega} = \frac{l_r C_{\alpha,r} - l_f C_{\alpha,f}}{I_z V_{lon}} V_{lat} - \frac{l_f^2 C_{\alpha,f} + l_r^2 C_{\alpha,r}}{I_z V_{lon}} + \frac{C_{\alpha,f}\, l_f}{I_z} \delta$$

This is equivalent to the following state space representation — B

states: $x_1 \triangleq V_{lat}$, $x_2 \triangleq \varphi$, $x_3 \triangleq \omega$ input: $u = \delta$

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \dot{V}_{lat} \\ \dot{\varphi} \\ \dot{\omega} \end{bmatrix} = \underbrace{\begin{bmatrix} -\frac{C_{\alpha,r} + C_{\alpha,f}}{m V_{lon}} & 0 & \frac{C_{\alpha,r} l_r - C_{\alpha,f} l_f}{m V_{lon}} - V_{lon} \\ 0 & 0 & 1 \\ \frac{l_r C_{\alpha,r} - l_f C_{\alpha,f}}{I_z V_{lon}} & 0 & \frac{-l_f^2 C_{\alpha,f} + l_r^2 C_{\alpha,r}}{I_z V_{lon}} \end{bmatrix}}_{\text{system matrix}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} \frac{C_{\alpha,f}}{m} \\ 0 \\ \frac{C_{\alpha,f}}{I_z} l_f \end{bmatrix}}_{\text{input matrix}} u$$

The Bicycle model

* with $\dot{x} = V_{lon} \cos\varphi - V_{lat} \sin\varphi$ & $\dot{y} = V_{lon} \sin\varphi + V_{lat} \cos\varphi$ the model can be augmented by the global position to a non linear state space model

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Kinematic bicycle model

System Modeling

- motion Update
- measurement Update

Reference frames:

$$x = \begin{pmatrix} x_v \\ y_v \\ \psi_v \end{pmatrix}$$

(x_v, y_v): global position of the car

ψ_v: heading angle

$$x = \begin{pmatrix} x_\ell \\ y_\ell \end{pmatrix}$$

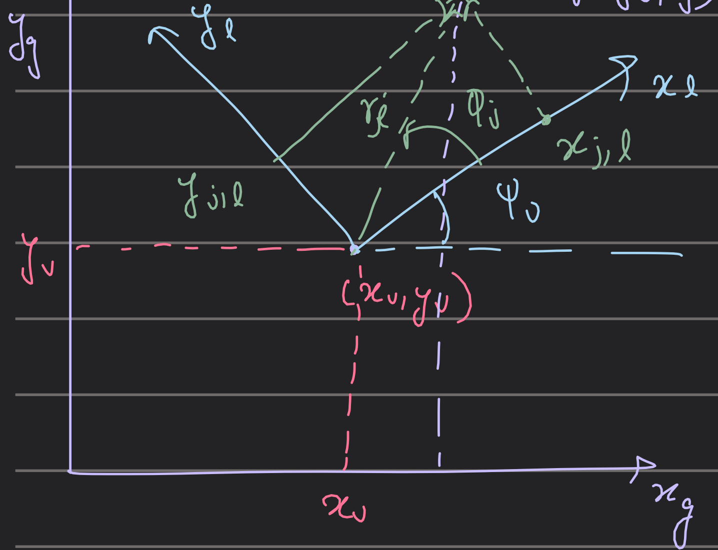
: (x_ℓ, y_ℓ): pose of the car wrt the local frame

the direction of x_ℓ is along the heading angle

the global coord of landmarks (cones): $\begin{pmatrix} x_{j,g} \\ y_{j,g} \end{pmatrix}$

the local coord of landmarks (cones) = $\begin{pmatrix} x_{j,\ell} \\ y_{j,\ell} \end{pmatrix}$

$(x_{j,g}, y_{j,g}) \longrightarrow$ a cone



$$\begin{pmatrix} x_{j,\ell} \\ y_{j,\ell} \end{pmatrix} = \begin{pmatrix} r_j \cos(\theta_j) \\ r_j \sin(\theta_j) \end{pmatrix}$$

$$\begin{pmatrix} r_j \\ \theta_j \end{pmatrix} = \begin{pmatrix} \sqrt{x_{j,\ell}^2 + y_{j,\ell}^2} \\ \tan^{-1}\left(y_{j,\ell} / x_{j,\ell} \right) \end{pmatrix}$$

$$\begin{pmatrix} r_j \\ \theta_j \end{pmatrix} = \begin{pmatrix} \sqrt{(x_{j,g} - x_v)^2 + (y_{j,g} - y_v)^2} \\ \operatorname{atan2}(y_{j,g} - y_v, x_{j,g} - x_v) - \psi_v \end{pmatrix}$$

Motion Model: used to compute the state transition & describe the robot's movement in the environment, given by control u_t,

$$\begin{pmatrix} x_{v,t} \\ y_{v,t} \\ \psi_{v,t} \end{pmatrix} = \begin{pmatrix} x_{v,t-1} \\ y_{v,t-1} \\ \psi_{v,t-1} \end{pmatrix} + \begin{pmatrix} \cos\psi_v & -\sin\psi_v & 0 \\ \sin\psi_v & \cos\psi_v & 0 \\ 0 & 0 & 1 \end{pmatrix} \Delta t$$

$$= \begin{pmatrix} x_{v,t-1} \\ y_{v,t-1} \\ \psi_{v,t-1} \end{pmatrix} + \begin{pmatrix} \cos\psi_v \Delta t & -\sin\psi_v \Delta t & 0 \\ \sin\psi_v \Delta t & \cos\psi_v \Delta t & 0 \\ 0 & 0 & \Delta t \end{pmatrix}$$

$$= g(u_t, x_{t-1}, 0)$$

the true velocity is approximated by measured velocity along with some error

$\hookrightarrow$ Gaussian noise with a covariance Q_t

$$\begin{pmatrix} x_{v,t} \\ y_{v,t} \\ \psi_{v,t} \end{pmatrix} = g(v_t, x_{t-1}) + \mathcal{N}(0, Q_t)$$

$\rightarrow$ from CLT the error is approximate as a Gaussian distribution

Using Taylor's eqⁿ: with $x_{0,t-1} = \mu_{t-1}$

$$g(v_t, x_{t-1}) = g(v_t, v_{t-1}) + G_t(x_{t-1} - \mu_{t-1})$$

$$G_t = \begin{pmatrix} 1 & 0 & (v_x \sin(\mu_{t-1}, \psi_0) - v_y \cos(\mu_{t-1}, \psi_0))\Delta t \\ 0 & 1 & (v_x \cos(\mu_{t-1}, \psi_0) - v_y \sin(\mu_{t-1}, \psi_0))\Delta t \\ 0 & 0 & \Delta t \end{pmatrix}$$

<u>Velocity Estimation</u>

<u>IMU</u>: $\dot{\psi}_0 \hookrightarrow$ yaw rate, $(v_x, v_y) = ?$ $\rightarrow$ measured using ω_{mot} & i_{gear} $\hookrightarrow$ gearing ratio

$$v_x = \frac{2\pi r_{wheel} (\omega^{rl}_{mot} + \omega^{rr}_{mot})/2}{i_{gear}} = \frac{\pi r_{wheel} (\omega^{rl}_{mot} + \omega^{rr}_{mot})}{i_{gear}}$$

ω^{rl}_{mot}: left wheel

ω^{rr}_{mot}: right wheel

Since the sensors measures longitudinal vel $\rightarrow$ assume: $v_y = 0$

$\downarrow$ assuming the lateral velocity to be low enough

$$\omega_{mot} = \frac{2\pi \, RPM}{60}$$

Measurement Model : → landmarks in cylindrical coordinates & there is some noise in the measurement model just like the motion model

$$\begin{pmatrix} r_j \\ \theta_j \end{pmatrix} = \begin{pmatrix} \sqrt{(x_{j,g}-x_v)^2 + (y_{j,g}-y_v)^2} \\ \operatorname{atan2}(y_{j,g}-y_v,\ x_{j,g}-x_v) - \varphi_v \end{pmatrix} + \mathcal{N}(0, R_t)$$

$$= h(x_t, j, m) + \mathcal{N}(0, R_t)$$

from the Taylor's approximation: with $x_0 = \mu$

$$h(x_t, j, m) \approx h(\mu_t, j, m) + H_t(x_t - \mu_t)$$

let

$$\delta = \begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} = \begin{pmatrix} \mu_{x_{j,g}} - \mu_{x_v} \\ \mu_{y_{j,g}} - \mu_{y_v} \end{pmatrix}$$

$q = \delta^T \delta$ → the Jacobian H_t is

$$H_t = \frac{\partial h(\mu_t, j, m)}{\partial v} = \begin{pmatrix} -\sqrt{q}\,\delta_x & -\sqrt{q}\,\delta_y & 0 \\ \delta_y & -\delta_x & -q \end{pmatrix}$$

$\mu_j = (\mu_{x_{j,g}}\ \mu_{y_{j,g}})^T$ → car's belief of the jth landmark's pose in the global coord

$\mu_v = (\mu_{x_v}, \mu_{y_v}, \mu_{\varphi_v})^T$

↓

is the internal belief of the vehicle's pose in the world frame

(*) In case of EKF SLAM, the state vector will contain the landmark pose. ∴ complete Jacobian contains zeros in all places except for the partial Jacobians $\frac{\partial h(\mu_t, j, m)}{\partial \mu_v}$ & $\frac{\partial h(\mu_t, j, m)}{\partial \mu_j}$

$$H_{t,j} = \frac{\partial h(\mu_t, j, m)}{\partial \mu_j} = \begin{pmatrix} \sqrt{q}\,\delta_x & \sqrt{q}\,\delta_y \\ -\delta_y & \delta_x \end{pmatrix}$$

Inverse measurement model : when a new landmark is initialized , the inv measurement model h^{-1} is needed

$$\begin{pmatrix} x_{j,g} \\ y_{j,g} \end{pmatrix} = \begin{pmatrix} x_v + r_j \cos(\psi_v + \theta_j) \\ y_v + r_j \sin(\psi_v + \theta_j) \end{pmatrix} + \mathcal{N}(0, R_t)$$

$$= h^{-1}(x_v, z_t) + \mathcal{N}(0, R_t)$$

The Taylor approximation wrt μ_v & the measurement z_j are

$$H_v^{-1} = H_v^{inv} = \frac{\partial h^{inv}(\mu_v, z_j)}{\partial \mu_v} = \begin{pmatrix} 1 & 0 & -r_j \sin(\mu_v + \theta_j) \\ 0 & 1 & r_j \cos(\mu_v + \theta_j) \end{pmatrix}$$

$$H_j^{-1} = H_j^{inv} = \frac{\partial h^{inv}(\mu_v, z_j)}{\partial \mu_j} = \begin{pmatrix} 1 & 0 & -r_j \sin(\mu_v + \theta_j) \\ 0 & 1 & r_j \cos(\mu_v + \theta_j) \end{pmatrix}$$

Handling Dropped Odometry Messages

Handling Dropped Odometry Messages

For various reasons, it may occur from time to time that an odometry message is not processed in time and the next message is therefore dropped. Generally, this is not a big issue. The pose estimate using the odometry data is a rough guess anyway, which is why the pose is corrected in the update step of EKF SLAM or the optimization of GraphSLAM.

If messages are dropped too frequently, however, the pose estimate may be too fraudulent to be corrected later on. Because of this, and to improve the short-term estimate (which is the one used by the controller), dropped messages are considered in the motion model by increasing the time step Δt depending on the number of dropped messages.

The motion model used here can be regarded as a numerical integration of the velocity estimates to form a pose estimate. Increasing the step width is therefore equivalent to reducing the resolution of the integration. While this obviously reduces the precision when compared to integrating with full resolution, the estimate is still more precise than it would be when just skipping a step of the integration.